

EduVision_DV: Dashboard Architecture & Storyboard

Milestone 2 / Module 4: UI/UX Specification, KPI Wireframes & Navigation Architecture

Project Statement: EduVision_DV is a unified, enterprise-grade Higher Education Intelligence Suite developed for Tableau Desktop. It integrates multi-source datasets—QS World University Rankings 2025, Times Higher Education 2024, World University Rankings 2023, and World Bank EdStats—to deliver four seamlessly interlinked dashboards with standardized master identifiers (University_ID, Country_ID, Year).

Dashboard Name	Target Analytical Audience	Primary Visual Elements	Interlinking Actions
1. University Overview	Executive Leadership, Applicants, Regents	6 KPI Cards, Global Rank Bar Chart, Regional Donut, Performance Scatter	Passes University_ID to Dashboards 2 & 3; Country_ID to Dashboard 4
2. Research Analytics	Provosts, Research Deans, Grant Agencies	Research Productivity Index, Citation vs Score, Top Research Hubs	Pre-filtered by selected University_ID with peer benchmark overlay
3. Student Analytics	Admissions Officers, International Recruiters	International Student %, Student-to-Staff Ratio, Gender Demographics	Pre-filtered by selected University_ID; cross-links to Country view
4. Country Comparison	Government Policymakers, Education Economists	National Higher Ed Capacity, Public Spend vs Output, Tertiary Enrollment	Receives Country_ID; allows benchmark comparison across 5 global regions

Visual Theme & Color Palette: Professional Dark Theme (#0F172A Slate / Navy Blue Background) with vibrant metric accents: Emerald Green (#10B981) for Academic Reputation, Cyan (#06B6D4) for Research Impact, Amber (#F59E0B) for International Students, and Purple (#8B5CF6) for Student Ratios. High contrast text (#F8FAFC) ensures WCAG AA accessibility across all display resolutions.

Dashboard 1: University Overview (Landing Page)

Provides high-level institutional scorecard and global performance benchmarking.

	KPI 4: Research Impact 100.0 / 100 Citations/Faculty

Dashboard 2: Research Analytics & Output

Deep-dive into research output, citation performance, and cross-border research networks.

	Research Productivity Index 95.4 / 100 Derived KPI (Sec 22)
MIT, and Caltech with citation impact benchmark lines.	
cross major university clusters.	

Dashboards 3 & 4: Student Analytics & Country Benchmarking

Student demographics, faculty staffing capacity, and country-level education environment.

Dashboard 3: Student Analytics (Demographics & Diversity)

Key Visualizations:

- **International Student % Distribution:** Histogram displaying international student proportions.
- **Student-to-Staff Ratio Benchmarking:** Bar chart comparing students per staff across institutions.
- **Gender Ratio Analysis:** Stacked bar breakdown of Female:Male ratios across institutions.
- **Total Student Enrollment vs International Share:** Bubble chart sizing enrollment.

Dashboard 4: Country Comparison (World Bank EdStats Benchmarking)

Key Visualizations:

- **National University Capacity:** Count of ranked world-class universities per nation.
- **Average National Ranking & Score:** Country-level aggregated institutional performance.
- **Government Education Spend (% GDP) vs Performance:** Scatter plot analyzing state investment vs top university density.
- **Tertiary Enrollment Rate Trends (2010–2023):** Historical multi-line trend comparison.

Interlinking Workflow & Filter Action Architecture

- Action 1 (University Drilldown):** Clicking any university on Dashboard 1 filters Dashboards 2 & 3 via `university_id`.
- Action 2 (Country Exploration):** Clicking a university's country on Dashboard 1 navigates to Dashboard 4 pre-filtered to that `country_id`.
- Action 3 (Navigation Hub):** Top navigation buttons on all dashboards enable seamless one-click switching with persistent filter states.